THE TOPOLOGY DEFINED BY THE COVERINGS OF A SITE HAS THE SAME SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The axioms of a site and of a topology. Same chapter, Definition definition-site and Definition definition-topology. Supplies the axioms matched against each other in the first part.
- The sieve generated by a family of morphisms. Same chapter, Definition definition-sieve-generated. Supplies the sieve appearing in the definition of the topology J.
- The sheaf conditions for a topology and for a site. Same chapter, Definition definition-sheaf-sets-topology and Definition definition-sheaf-sets. Supplies the two conditions compared in the second part.

Source reference. Original definition 00Z1: sources/stacks-sites/pinned-original-source.tex, source lines 11468–11468. Navigation pointer invoked at proof line 11468; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 00Z8: sources/stacks-sites/pinned-original-source.tex, source lines 11472–11472. Navigation pointer invoked at proof line 11472; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 00VM: sources/stacks-sites/pinned-original-source.tex, source lines 11474–11474. Navigation pointer invoked at proof line 11474; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 00VH: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 526–526. Navigation pointer invoked at proof line 11481; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original definition 00Z4: sources/stacks-sites/pinned-original-source.tex, source lines 11484–11484. Navigation pointer invoked at proof line 11484; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site with coverings $Cov(\mathcal{C})$. For every object U of $\mathcal{C}$, let $J(U)$ denote the set of sieves S on U with the following property: there exists a covering $\{f_i : U_i \to U\}_{i\in I} \in Cov(\mathcal{C})$ so that the sieve S' generated by the f_i (see Definition 00Z1) is contained in S.*

1. *This J is a topology on $\mathcal{C}$.*
2. *A presheaf $\mathcal{F}$ is a sheaf for this topology (see Definition 00Z8) if and only if it is a sheaf on the site (see Definition 00VM).*

Proof. To prove the first assertion we just note that axioms (1), (2) and (3) of the definition of a site (Definition 00VH) directly imply the axioms (3), (2) and (1) of the definition of a topology (Definition 00Z4). As an example we prove J has property (2). Namely, let U be an object of $\mathcal{C}$, let S, S' be sieves on U such that $S \in J(U)$, and such that for every $V \to U$ in $S(V)$ we have $S' \times_U V \in J(V)$. By definition of $J(U)$ we can find a covering $\{f_i : U_i \to U\}$ of the site such that S the image of $h_{U_i} \to h_U$ is contained in S. Since each $S' \times_U U_i$ is in $J(U_i)$ we see that there are coverings $\{U_{ij} \to U_i\}$ of the site such that $h_{U_{ij}} \to h_{U_i}$ is contained in $S' \times_U U_i$. By definition of the base change this means that $h_{U_{ij}} \to h_U$ is contained in the subpresheaf $S' \subset h_U$. By axiom (2) for sites we see that $\{U_{ij} \to U\}$ is a covering of U and we conclude that $S' \in J(U)$ by definition of J.

Let $\mathcal{F}$ be a presheaf. Suppose that $\mathcal{F}$ is a sheaf in the topology J. We will show that $\mathcal{F}$ is a sheaf on the site as well. Let $\{f_i : U_i \to U\}_{i\in I}$ be a covering of the site. Let $s_i \in \mathcal{F}(U_i)$ be a family of sections such that $s_i|_{U_i\times_U U_j} = s_j|_{U_i\times_U U_j}$ for all i, j. We have to show that there exists a unique section $s \in \mathcal{F}(U)$ restricting back to the s_i on the U_i. Let $S \subset h_U$ be the sieve generated by the f_i. Note that $S \in J(U)$ by definition. In stead of constructing s, by the sheaf condition in the topology, it suffices to construct an element

$$\varphi \in \mathrm{Mor}_{PSh(\mathcal{C})}(S, \mathcal{F}).$$

Take $\alpha \in S(T)$ for some object $T \in \mathcal{U}$. This means exactly that $\alpha : T \to U$ is a morphism which factors through f_i for some $i \in I$ (and maybe more than 1). Pick such an index i and a factorization $\alpha = f_i \circ \alpha_i$. Define $\varphi(\alpha) = \alpha_i^* s_i$. If i', $\alpha = f_i \circ \alpha'_{i'}$ is a second choice, then $\alpha_i^* s_i = (\alpha'_{i'})^* s_{i'}$ exactly because of our condition $s_i|_{U_i\times_U U_j} = s_j|_{U_i\times_U U_j}$ for all i, j. Thus $\varphi(\alpha)$ is well defined. We leave it to the reader to verify that φ, which in turn determines s is correct in the sense that s restricts back to s_i.

Let $\mathcal{F}$ be a presheaf. Suppose that $\mathcal{F}$ is a sheaf on the site $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$. We will show that $\mathcal{F}$ is a sheaf for the topology J as well. Let U be an object of $\mathcal{C}$. Let S be a covering sieve on U with respect to the topology J. Let

$$\varphi \in \mathrm{Mor}_{PSh(\mathcal{C})}(S, \mathcal{F}).$$

We have to show there is a unique element in $\mathcal{F}(U) = \mathrm{Mor}_{PSh(\mathcal{C})}(h_U, \mathcal{F})$ which restricts back to φ. By definition there exists a covering $\{f_i : U_i \to U\}_{i \in I} \in \mathrm{Cov}(\mathcal{C})$ such that $f_i : U_i \to U$ belongs to $S(U_i)$. Hence we can set $s_i = \varphi(f_i) \in \mathcal{F}(U_i)$. Then it is a pleasant exercise to see that $s_i|_{U_i \times_U U_j} = s_j|_{U_i \times_U U_j}$ for all i, j. Thus we obtain the desired section s by the sheaf condition for $\mathcal{F}$ on the site $(\mathcal{C}, \mathrm{Cov}(\mathcal{C}))$. Details left to the reader. $\square$